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CHARACTERISING BIAS IN COMPRESSED MODELS

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ABSTRACT

The popularity and widespread use of pruning and quantization is driven by the severe resource constraints of deploying deep neural networks to environments with strict latency, memory and energy requirements. These techniques achieve high levels of compression with negligible impact on top-line metrics (top-1 and top-5 accuracy). However, overall accuracy hides disproportionately high errors on a small subset of examples; we call this subset Compression Identified Exemplars (*CIE*). We further establish that for *CIE* examples, compression amplifies existing algorithmic bias. Pruning disproportionately impacts performance on underrepresented features, which often coincides with considerations of fairness. Given that *CIE* is a relatively small subset but a great contributor of error in the model, we propose its use as a human-in-the-loop auditing tool to surface a tractable subset of the dataset for further inspection or annotation by a domain expert. We provide qualitative and quantitative support that *CIE* surfaces the most challenging examples in the data distribution for human-in-the-loop auditing.

1 Introduction

Pruning and quantization are widely applied techniques for compressing deep neural networks, often driven by the resource constraints of deploying models to mobile phones or embedded devices (Esteva et al., 2017; Lane & Warden, 2018). To-date, discussion around the relative merits of different compression methods has centered on the trade-off between level of compression and top-line metrics such as top-1 and top-5 accuracy (Blalock et al., 2020). Along this dimension, compression techniques are remarkably successful. It is possible to prune the majority of weights (Gale et al., 2019; Evci et al., 2019) or heavily quantize the bit representation (Jacob et al., 2017) with negligible decreases to test-set accuracy.

However, recent work by Hooker et al. (2019b) has found that the minimal changes to top-line metrics obscure critical differences in generalization between pruned and non-pruned networks. The authors establish that pruning disproportionately impacts predictive performance on a small subset of the dataset. We build upon this work and focus on the implications of these findings for a dataset with sensitive protected attributes such as gender and age. Our work addresses the question: *Does compression amplify existing algorithmic bias?*

Understanding the relationship between compression and algorithmic bias is particularly urgent given the widespread use of compressed deep neural networks in resource constrained but sensitive domains such as hiring (Dastin, 2018; Harwell, 2019), health care diagnostics (Xie et al., 2019; Gruetzemacher et al., 2018; Badgeley et al., 2019; Oakden-Rayner et al., 2019), self-driving cars (NHTSA, 2017) and facial recognition software (Buolamwini & Gebru, 2018b). For these tasks, the trade-offs incurred by compression may be intolerable given the impact on human welfare.

We establish consistent results across widely used quantization and pruning techniques and find that compression amplifies algorithmic bias. The minimal changes to overall accuracy hide disproportionately high errors on a small subset of examples. We call this subset Compression Identified Exemplars (*CIE*). Given two model populations, one

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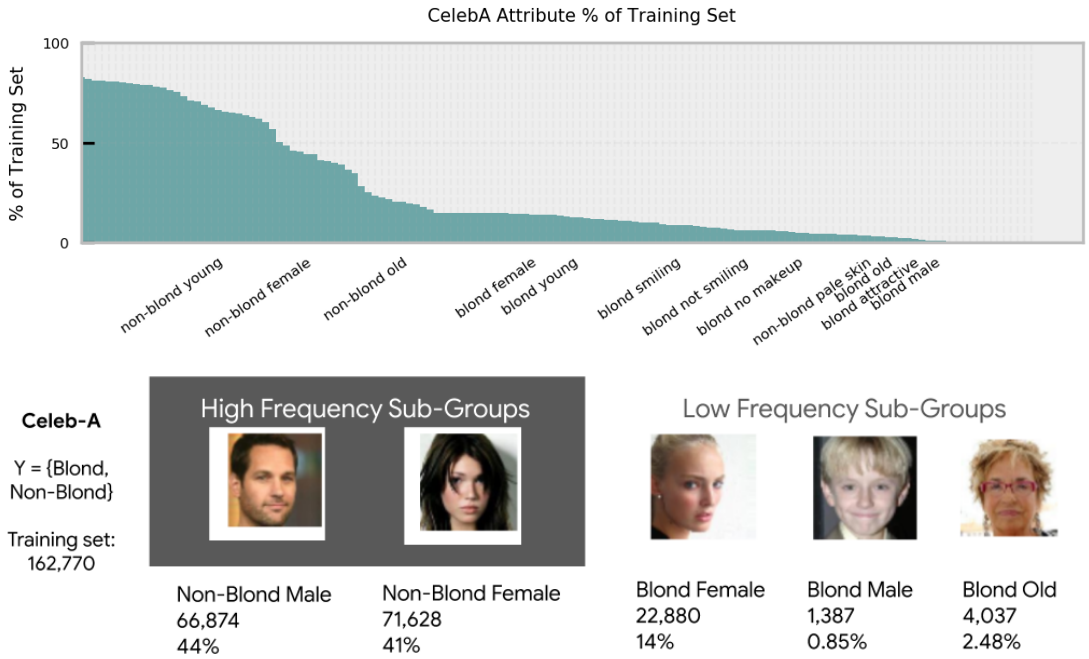

Figure 1: Most natural image datasets exhibit a long-tail distribution with an unequal frequency of attributes in the training data. Below each attribute sub-group in CelebA, we report the share of training set and total frequency count.

compressed and one non-compressed, an example is a *CIE* if the labels predicted by the compressed population diverges from the labels produced by the non-compressed population.

Reasoning about model behavior is often easier when presented with a subset of data points that is atypical or hard for the model to classify. Our work proposes *CIE* as a method to surface a tractable subset of the dataset for auditing. One of the biggest bottlenecks for human auditing is the large scale size of modern datasets and the cost of annotating each feature (Veale & Binns, 2017). For many real-world datasets, labels for protected attributes are not available. In this paper, we show that *CIE* is able to automatically surface more challenging examples and over-indexes on the protected attributes which are disproportionately impacted by compression. *CIE* is a powerful unsupervised protocol for auditing. Given that the methodology is agnostic to the presence of attribute labels, *CIE* allows us to audit multiple attributes all at once. This makes *CIE* a potentially valuable human-in-the-loop auditing tool for domain experts when labels for underlying attributes are limited.

In Section. 2, we firstly establish the degree to which model compression amplifies forms of algorithmic bias using traditional error metrics. Section. 3 introduces different measures of *CIE* and motivates the use of *CIE* as an auditing tool for surfacing these biases when labels are not available for the underlying protected attributes. In Section. 3.2 we discuss a human-in-the-loop protocol to audit compression induced error.

2 Characterising Compression Induced Bias in Data with Sensitive Attributes

Recent studies have exposed the prevalence of undesirable biases in machine learning datasets. For example, Buolamwini & Gebru (2018a) discuss the disparate treatment of darker skin tones due to under-representation within facial analysis datasets, object detection datasets tend to under-represent images from lower income and non-Western regions (Shankar et al., 2017; DeVries et al., 2019), activity recognition datasets exhibit stereotype-aligned gender biases (Zhao et al., 2017), and word co-occurrences within text datasets frequently reflect social biases relating to gender, race and disability (Garg et al., 2017; Hutchinson et al., 2020).

In the absence of fairness-informed interventions, trained models invariably reflect the undesirable biases of the data they are trained on. This can result in higher overall error rates on demographic groups underrepresented across the entire dataset and/or false positive rates and false negative rates that skew in alignment with the over- or under-representation of demographic groups *within* a target label.

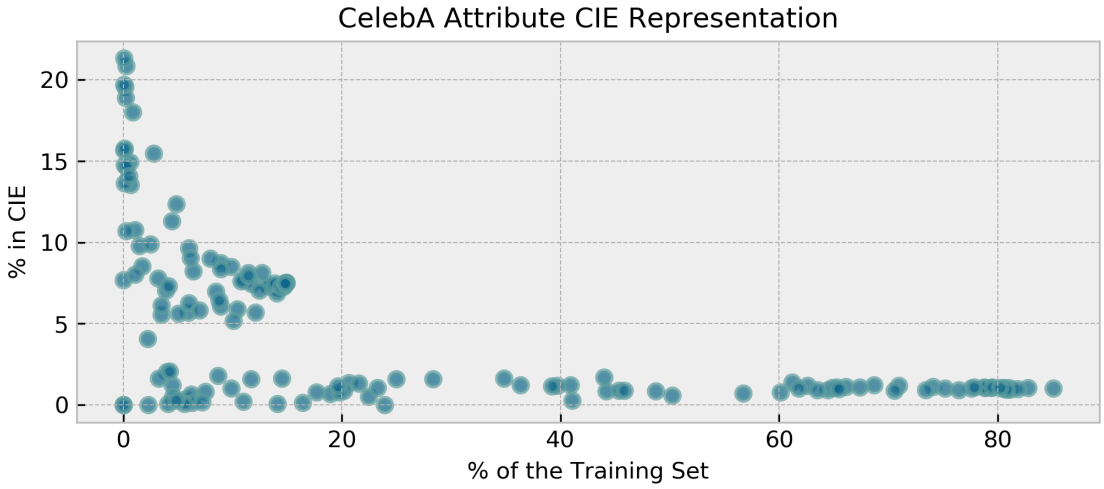

Figure 2: Plot of the fraction of the training set of each attribute in CelebA against the relative representation of each attribute in CIE_p. CIE_p over-index on underrepresented attributes in the dataset. In this plot we threshold Taxicab *CIE* generated from a pruned model at 80%.

In this section, we firstly establish the degree to which model compression amplifies forms of algorithmic bias using traditional error metrics. Our analysis leverages CelebA (Liu et al., 2015), a dataset of celebrity faces annotated with 40 binary face attributes and trains a classifier to predict a binary label indicating if the `Blonde hair` attribute is present. The CelebA dataset is well-suited for our analysis due to the significant correlations between protected demographic groups and the target label (defined by `Blonde`), as well as the overall under-representation of some demographic groups across the training dataset. As seen in Figure 1, CelebA is representative of many natural image datasets where attributes follow a long-tail distribution (Zhu et al., 2014; Feldman, 2019).

2.1 Methodology

Our goal is to understand the implications of compression on model bias and fairness considerations. Thus, we focus attention on two protected unitary attributes `Male` and `Young` and one intersectional attribute from the combination of these unitary attributes (i.e `Young Male`). To characterize the impact of compression on age and gender sub-groups we compare sub-group error rate, false positive rate (FPR) and false negative rate (FNR) between a baseline (i.e. non-compressed) and models pruned and quantized to different levels of compression (i.e. compressed).

We evaluate three different compression approaches: magnitude pruning (Zhu & Gupta, 2017), fixed point 8-bit quantization (Jacob et al., 2017) and hybrid 8-bit quantization with dynamic range (Williamson, 1991). In contrast to the pruning which is applied progressively over the course of training, all of the quantization methods we evaluate are implemented post-training. For all experiments, we train a ResNet-18 (He et al., 2015) on CelebA for $10,000$ steps with a batch size of 256.

Pruning Protocol For pruning, we vary the end sparsity for $t \in \{0.3, 0.5, 0.7, 0.9, 0.95, 0.99\}$. For example, $t = 0.9$ indicates that 90% of model weights are removed over the course of training, leaving a maximum of 10% non-zero weights at inference time. For the pruning variants, we prune every 500 steps between 1000 and 9000 steps. These hyperparameter choices were based upon a limited grid search which suggested that these particular settings minimized degradation to test-set accuracy across all pruning levels. At the end of training, the final pruned mask is fixed and during inference only the remaining weights contribute to the model prediction. To move beyond anecdotal observations, we train 30 models for every level of compression considered. Our goal is to have a high level of certainty that differences in predictive performance between compressed and non-compressed models is statistically significant and not due to inherent noise in the stochastic training process of deep neural networks.

Quantization Protocol We use two types of post-training quantization. The first type uses a hybrid "dynamic range" approach with 8-bit weights (Alvarez et al., 2016). The second type uses fixed-point only 8-bit weights (Vanhoucke